

SUMANTH YALAMARTY

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Email ♦ Website ♦ Github ♦ Twitter ♦ Podcast

EDUCATION

Bachelors of Computing, National University of Singapore (*GPA: 4/5*) Aug 2020 to Jun 2024
Major in **Computer Science** (*Honors w/ Distinction*), Focus Area in **Artificial Intelligence** and **Software Engineering**

EXPERIENCE

Quantitative Strategist July 2024 to Present
Goldman Sachs *Singapore, SG*

- Build on Atlas (firm's low-latency trading execution system) in **Global Banking & Markets**
- Shipped **Lit/Dark/Close SOR** for **7 markets** across **25+ venues**; productionized routing, venue-microstructure logic
- Lifted internalization hit rates by **35%** via liquidity caching and flow segmentation/binning
- Migrated **1.5bn+\$/day** notional from legacy to Atlas within **2 weeks**, without any incidents

Software Engineering Intern May 2023 to Aug 2023

- Worked on SecDB, part of the **Core Engineering** division
- Built end-to-end observability for an ingestion pipeline processing **10bn+ transactions/day** across **20+ DB rings**
- Delivered to production in **6 weeks**; only intern to ship; design still runs as of today

Research Intern Jan 2022 to Aug 2022, Aug 2023 to Jan 2024

NUS IORA (Institute of Operations Research and Analytics) *Singapore, SG*

- POC delivered in 4 weeks to use **federated learning** techniques to improve point of sale and loyalty systems
- Built a fact-checking system using LLMs, vector databases and knowledge graph grounding techniques

TEACHING

Undergraduate Teaching Assistant Multiple Terms (Jan 2022 - May 2024)

NUS School of Computing *Singapore*

- Teaching assistant: **CS3243**, **IT5005** (Artificial Intelligence), **CS1010**, **CS2030S** (Programming Methodology), **CS3203** (Software Engineering), **CS2102** (Database Systems) with a rating of **4.5/5**; **Honor's List of Student Tutors**

Senior Advisor May 2023 to Aug 2023, May 2024 to Aug 2024

- Advised **24** student teams on a **self-directed software development** journey throughout the summer

PROJECTS

Toy Sequencer | *C++*, *Protobuf*, *CMake* [Github](#)

Multi producer, multi consumer queue mimicking high speed trading systems

Q-Indexer | *KDB+*, *q*, *WASM*, *TypeScript* [Github](#)

Shared indexer for kdb+ queries used internally by the trading desk and strats

QuizMe | *AI*, *LLMs* [Github](#)

AI generates questions and UI on the fly to test you in programming languages

Perzona | *T3 Stack*, *Fal.ai*, *AWS*, *Inngest* [Github](#)

Lip synced videos using AI using either text or audio

NUS Learn AI, SELENE | *Python*, *LangChain*, *Large Language Models*, *Neural QG* [Report](#), [Presentation](#), [Github](#)

Suite of **AI-enabled tools** to improve **NLP education** as a part of **FYP (Disseration)**

Static Program Analyzer for SIMPLE | *C++*, *CMake*, *Docusaurus* [Github](#)

Program analyzer for a custom language called **SIMPLE** supporting **semantic querying** of entity relationships

99.84% accuracy in evaluation with zero performance lapses, Best in the cohort

EXTRA-CURRICULAR ACTIVITIES

- **Blogger** about ideas from life
- **Polkadot Blockchain Academy** Graduate
- Active Community Member of the **Network States** and **Polkadot Singapore**
- **Student Wellness Ambassador** at Prince Georges' Park Residences

SKILLS

Technical Skills Python, Zero GC Java, Rust, Q/kdb+, C++, Go, TypeScript, $\text{\LaTeX}$, SQL, Terraform

Frameworks and Tools Langchain, PyTorch, OpenCode, Next, Expo, Rx, Effect, Fastify, Grafana, Prometheus